

Delayed Payment Interest

DPIC v1 · Release & QA Guide

Loan Management System — Accounting | Business & Integration Reference
Audience: QA · Collections · Payments · Product Ops

Version	1.0	Module	novopay-platform-accounting-v2
Date	12-Jun-2026	Branch	feature/delayed_payment_interest
Scope	DPI v1 — monthly frequency	Status	Build green · pending QA certification

This guide explains, in business language, what the DPI feature does, the new database objects and batch jobs, the new API response fields downstream teams integrate against, the environment prerequisites that must be configured before DPI works, and the QA test checklist. No source code is included.

Status legend:

DONE

live

PENDING

not built

AUTO

auto-applied (Flyway)

MANUAL

manual setup

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1 What is Delayed Payment Interest?

The business definition and the rules that govern DPI.

Delayed Payment Interest (DPI) is a **fifth loan charge** — alongside Principal, Interest, Penalty and Fee — levied on a customer's **overdue (Principal + Interest)** once an EMI crosses its grace window. It compensates the lender for delayed repayment, distinct from penal interest.

Rule	Behaviour
What it accrues on	Overdue Principal + Interest only. Never on outstanding DPI itself — there is no compounding.
Rate & grace	The loan's effective interest rate (ROI), with grace days and day-count (e.g. 360/365) configured per product scheme & frequency.
When it bills	DPI accrued for a missed EMI is billed on the next EMI due date — or, post-maturity, the next monthly anchor.
Appropriation order	On repayment DPI is settled after Interest, before Penalty/Fee (configurable; default position 3).
Knock-off / DPD	DPI residual does not block EMI knock-off; overdue DPI raises DPD automatically.
Scope (v1)	Monthly repayment frequency only.

2 How DPI flows

DPI runs through the same shape as normal interest, end to end.

Stage	What happens	Status
1 · Accrual	Daily, DPI is computed on overdue (P+I) for active monthly DPI loans and staged (no GL yet).	DONE
2 · Booking	The accrued DPI is posted to the General Ledger.	DONE
3 · Billing	Accrued DPI moves into customer dues on the next EMI — or post-maturity, the next monthly anchor.	DONE
4 · Appropriation	On repayment DPI is settled in its slot; paid DPI is surfaced on payment records.	DONE
5 · NPA movement	DPI moves to/from suspense in lock-step with interest across NPA.	DONE
6 · Lifecycle	Restructuring, foreclosure and post-maturity handled (see §7).	DONE
7 · Reversal	A reversed repayment carries its DPI portion back.	DONE

3 New batch jobs (End-of-Day)

Three EOD jobs to schedule for the tenant — they run in order, daily.

Job (Request)	Runs	What it does
<code>dpiAccrualCalculation</code>	EOD · 1st	Computes DPI on overdue (P+I) for active, monthly, DPI-enabled loans; writes accrual rows. No GL.
<code>dpiAccrualBooking</code>	EOD · 2nd	Posts the GL accrual leg for newly accrued DPI.
<code>dpiBilling</code>	EOD · 3rd	Moves accrued DPI into customer dues (component DPI) on the next EMI / post-maturity anchor.

NOTE All three are idempotent (windowed, with same-day anti-replay), so a re-run never double-posts. They must be scheduled in the order above, and registered in the API registry — see §4 (V000450) and §6.

4 New database objects

Created automatically by Flyway on deploy — no manual DB action.

Object	Type	Purpose	Flyway
<code>dpi_accrual_details</code>	New table	DPI accrual staging — the accrued → booked → billed lifecycle per loan.	V000187
<code>product_scheme_frequency_details</code>	New table	Per-frequency DPI config (applicability, grace, day-count).	V000190
<code>loan_account_payments_details.dpi_amount</code>	New column	DPI portion of each repayment.	V000188
<code>transaction_reversal_details.dpi_amount</code>	New column	DPI portion of a reversal.	V000188
<code>prepayment_details (billed_dpi_*, bpd_*)</code>	New columns	DPI waiver buckets for foreclosure / part-prepayment.	V000193
<code>loan_product_asset_criteria.sequence_5</code>	New columns	Appropriation slot for DPI (+ per-slot order).	V000194
Code masters: <code>DPI_APBL</code> , <code>APP_LOGIC_DPI</code>	Seed data	DPI on/off flag + DPI appropriation component (position 5).	V000119
api_master: the 3 DPI batch-job APIs	Seed data	Registers <code>dpiAccrualCalculation</code> , <code>dpiAccrualBooking</code> , <code>dpiBilling</code> in the platform API registry (service ACCOUNTING).	V000450

TIP DPI rows are identified by `component_type = 'DPI'` in `loan_due_details`.

5 Consumer & integration guide

New API response fields downstream teams integrate against. Amounts are strings; field names differ per API.

5.1 getLoanAccountOverviewDetails — Loan account overview DONE Primary integration point for Collections / Payments.

Response field	Meaning
dpi_overdue_amount	DPI past its due date.
dpi_due_amount	DPI currently due (not yet overdue).
dpi_paid_amount	DPI paid to date.
dpi_waived_amount	DPI waived to date.
dpi_amount	DPI split inside each payment_details_list[] entry.

NOTE The overview aggregates total_overdue / total_due now **include DPI** — those totals will be larger than before.

5.2 getLoanAccountSummaryDetails — Loan 360 summary DONE

Response field	Meaning
dpi_accrued_amount	DPI accrued but not yet billed.
dpi_current_due_amount	DPI currently due (different key than overview's dpi_due_amount).
dpi_overdue_amount	DPI past due date.
dpi_paid_amount / dpi_waived_amount	DPI paid / waived to date.
dpi_written_off_amount	DPI written off — always 0.00 today.

5.3 Repayment, statement, reversal & collection push DONE

API	Field	Meaning
loanRepayment	dpi_amount	DPI settled in this repayment (also stored on the payment record).
getLoanAccountStatement	dpi_amount	DPI split per transaction row (present when > 0).
loanAccountTransactionReversal	dpi_amount	DPI portion reversed.
recurringPayment (collection push)	dpi_due, dpi_overdue	DPI in the collection-to-payments payload; Total Outstanding, Total Overdue and Collection Amount all include DPI.

5.4 Foreclosure & part-prepayment previews DONE

API	Field(s)	Meaning
fetchLoanForeclosureSimulationDetails	billed_dpi, bpd_amount	Billed DPI + broken-period (unbilled accrued) DPI at the foreclosure cut-off.
getLoanForeclosureDetails	billed_dpi_details{}, bpd_details{}	Nested DPI buckets with waiver structure (also a request input on execute).

API	Field(s)	Meaning
loanPrepayment (foreclosure)	dpi_till_date_of_foreclosure	Unbilled DPI to foreclosure date, added to amount-to-collect.
getLoanAccountPartPrepaymentDetails	bpd_amount	Unbilled DPI to the part-prepayment date.

5.5 Product scheme configuration APIs DONE

API	Field(s)	Meaning
create / update / getProductSchemeDetails	interest_and_dpi_list[], dpi_applicable, dpi_applicable_value	Per-frequency interest + DPI configuration (DPI on/off and label).

5.6 No DPI field (by design) OUT OF SCOPE

Area	Note for consumers
Repayment Schedule API · getLoanAccountDetails · getLoanAccountBillingDetails	No DPI field — DPI is not a schedule line; it surfaces via overview / summary / collection push / dues only.

6 Environment pre-requisites

What must be in place before DPI works. AUTO items apply on deploy; MANUAL items are set up before testing.

6.1 Auto-applied on deploy AUTO

The new tables, columns, code masters and API registrations in §4 (V000119, V000187, V000188, V000190, V000193, V000194, V000450). No action — QA only confirms they exist after deployment.

6.2 Manual setup required before DPI functions MANUAL

Item	Owner	What & why
6 DPI General Ledgers	Product Ops (webapp)	DPI AIR, DPI income, billed-DPI receivable, billed-DPI waive, DPI suspense AIR, DPI suspense — one set per portfolio (JLGD / INDL / SHGD). Must exist before any DPI rule resolves.
~28 accounting rules	Product Ops / Vasanthi (webapp)	Map each DPI leg to its GLs. Accrual & billing rules must be is_reversible = FALSE. Each rule's source-amount must equal the reference code below.
6 transaction catalogue + rule rows	Vasanthi	The (type, sub-type) literals the engine posts (§6.3). Without them DPI postings fail to resolve.
product_scheme_frequency_details rows	Product / Rohit	Per scheme: dpi_applicable='YES', interest_frequency='MONTHLY', grace, day-count. The hard gate — no row = no DPI.
loan_product_asset_criteria.sequence_5	Data setup	Set to APP_LOGIC_DPI per product to pin the appropriation slot; else DPI falls back to position 3.

Item	Owner	What & why
Schedule the 3 EOD jobs	Env / Ops	dpiAccrualCalculation → dpiAccrualBooking → dpiBilling, in order, daily.
DPI go-live-date gating	Bank Ops (optional)	Not seeded — create manually only if go-live-date gating is required.

6.3 Transaction (type, sub-type) literals to seed

Transaction type	Sub-type	Used for	Rev.
INTEREST	DPI_NORMAL_ACCRUAL	Normal accrual booking	No
INTEREST	DPI_NPA_ACCRUAL	NPA accrual booking	No
BILLING	DPI_NORMAL_BILLING	Normal billing	No
INTEREST	DPI_NPA_ACCRUAL_BOOKING	NPA billing	No
REGULAR_TO_NPA	DPI_INT_INCOME	Move DPI income to suspense	—
NPA_TO_REGULAR	DPI_INT_INCOME	Move DPI income from suspense	—

Reference codes for rule source-amounts:

DPI_ACCR_INT_AMT · DPI_ACCR_INT_AMT_NPA · BILLED_DPI_INT_AMT · ADV_BILLED_DPI_INT_AMT ·
 DPI_INT_SUSP_AIR_AMT · DPI_ACCR_INT_SUSPENSE_AIR_AMT · UNPAID_BILLED_DPI_INT_AMT ·
 PAID_BILLED_DPI_INT_AMT · BILLED_DPI_INT_WAIVED_AMT · DPI_AMT

7 QA test checklist

What to verify in the deployment environment once §6 prerequisites are configured.

Area	Expected behaviour to verify	Status
Accrual	An active monthly DPI loan that goes overdue accrues DPI daily on overdue (P+I) after grace; never on DPI itself.	DONE
Booking & billing	Accrued DPI is booked to GL and billed on the next EMI date into dues (component DPI).	DONE
Appropriation	A repayment settles DPI after Interest and before Penalty/Fee; paid DPI is surfaced.	DONE
NPA movement	DPI moves to/from suspense together with interest across NPA tagging.	DONE
DPD	Overdue DPI raises DPD with no extra configuration.	DONE
Restructuring	Billed DPI preserved as overdue dues; accrued-unbilled DPI capitalised onto the first new installment; no waiver, no upfront, no double-bill.	DONE
Post-maturity	DPI keeps accruing after maturity while overdue; bills on the next monthly anchor; stops when overdue clears.	DONE

Area	Expected behaviour to verify	Status
Foreclosure / part-prepay	DPI shown in previews; unbilled DPI force-billed at closure.	DONE
Loan 360 / overview	All DPI response fields (§5) populate correctly.	DONE
Collection push	Collection-to-payments payload carries DPI (<code>dpi_due</code> , <code>dpi_overdue</code>) and includes it in Total Outstanding / Overdue / Collection Amount.	DONE
Reversal	A reversed repayment carries its DPI portion back.	DONE
DPI write-off	Dedicated DPI write-off path — not built in v1.	PENDING
Weekly / other frequency	v1 supports monthly only.	OUT OF SCOPE

8 Known gaps & out of scope (v1)

Item	Note
Collections-app DPI column (UI)	Accounting payload already carries DPI; rendering the column on the collection-app screen is a collection-app / webapp change (other team).
DPI write-off path	Not built; <code>dpi_written_off_amount</code> is always 0.00.
Weekly / bi-weekly / daily DPI	v1 is monthly only.
GL & accounting-rule seeding	Hard dependency — DPI postings will fail until the §6 GLs and rules are configured.

Prepared for QA certification of DPI v1. Build is green; status remains pending QA sign-off in the target environment.